

Medical Lab Management System

Allied Diagnostic



Group: WALP_3

Table of content

01. Group members

02. Introduction

03. Objectives

04. Functional requirements

05. Non-functional requirements

06. Architecture

07. Use cases

08. Demonstration

09. Trello board

10. GIT

11. Challenges

12. Final thoughts

Our team

M. A. Wijesinghe

723597915

Project Manager
Backend Developer

A. P. Paramananda

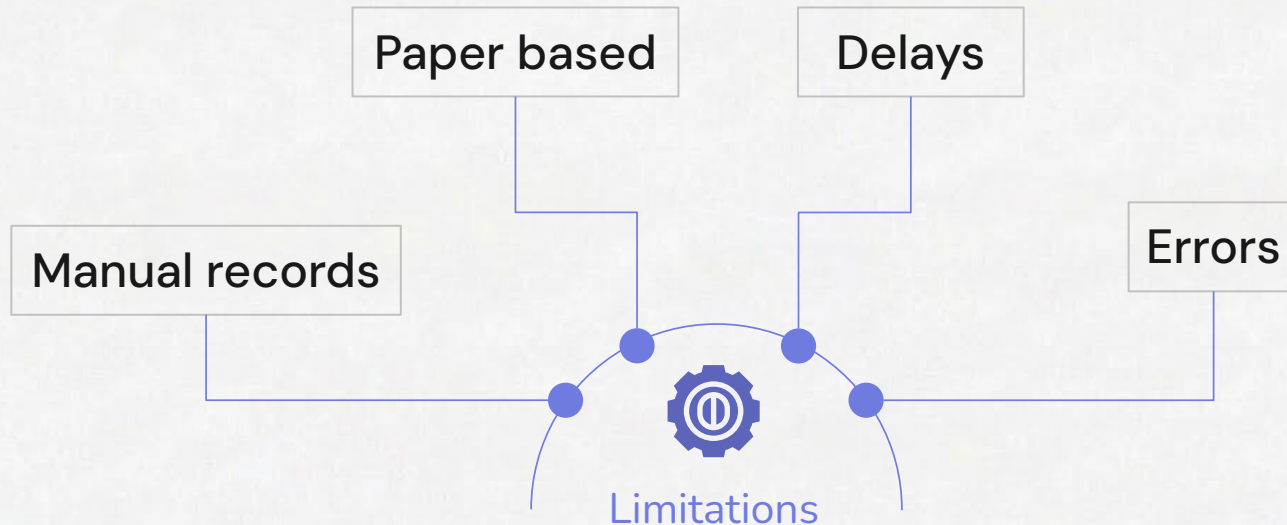
521420912

Frontend Developer
Tester



Introduction

Medical Lab Management System – Allied Diagnostics



Objectives

Patient

- Register Securely
- Book Tests
- Access Reports Online

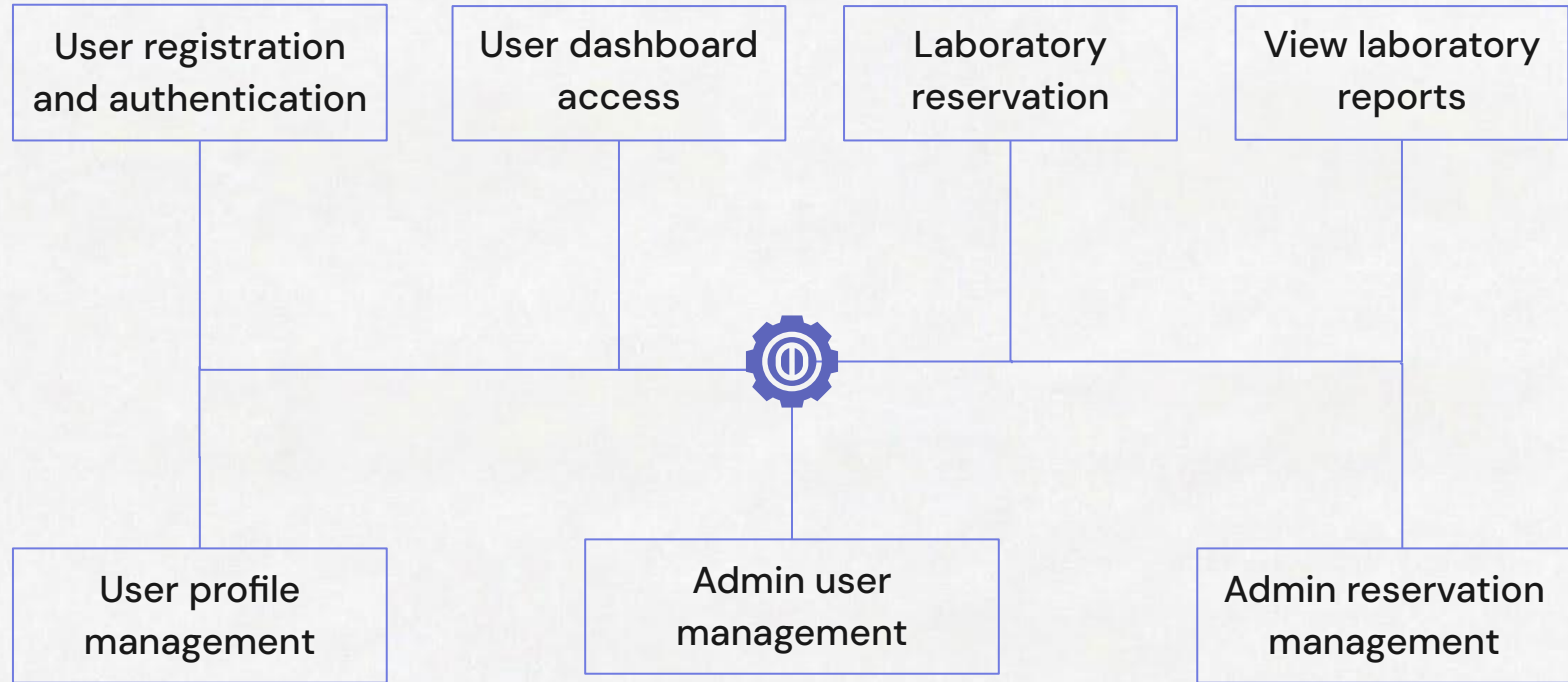


MLT

- Manage Patient Records
- Track Laboratory Tests
- Manage Users, Tests



Functional requirements



Non-functional requirements

Performance

- Page load time $\leq 3s$
- Support 100+ concurrent users

Reliability

- System availability $\geq 99\%$ uptime

Security

- User authentication required
- Role based access



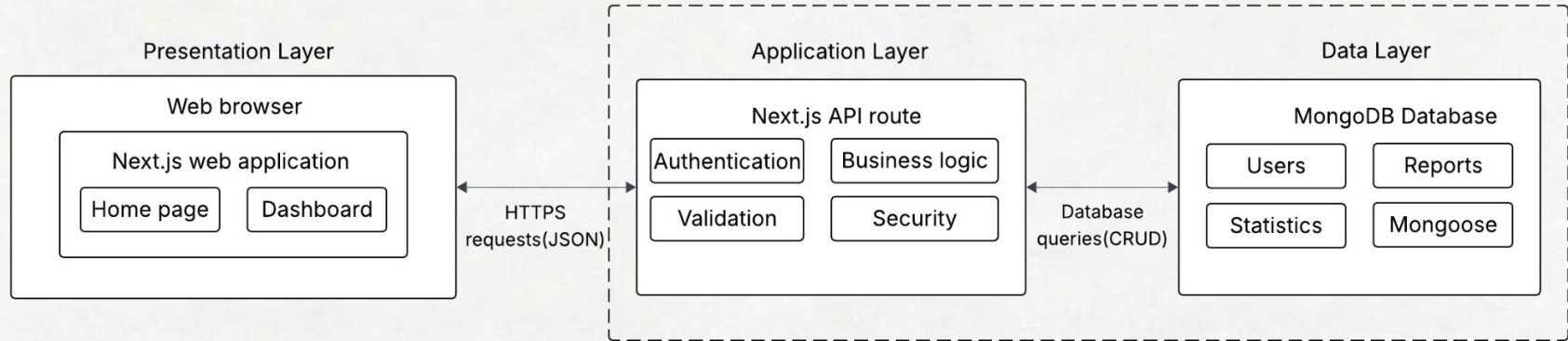
Usability

- Test booking $< 2min$
- Clear navigation and simple forms

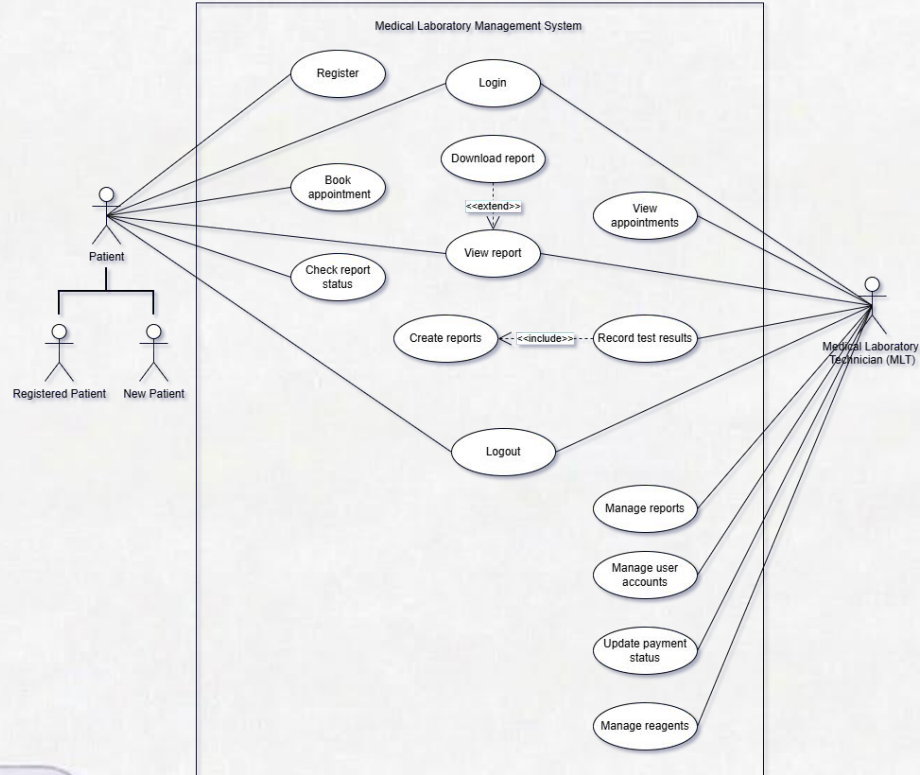
Scalability

- Handle high users load without performance drop
- Scalable database design

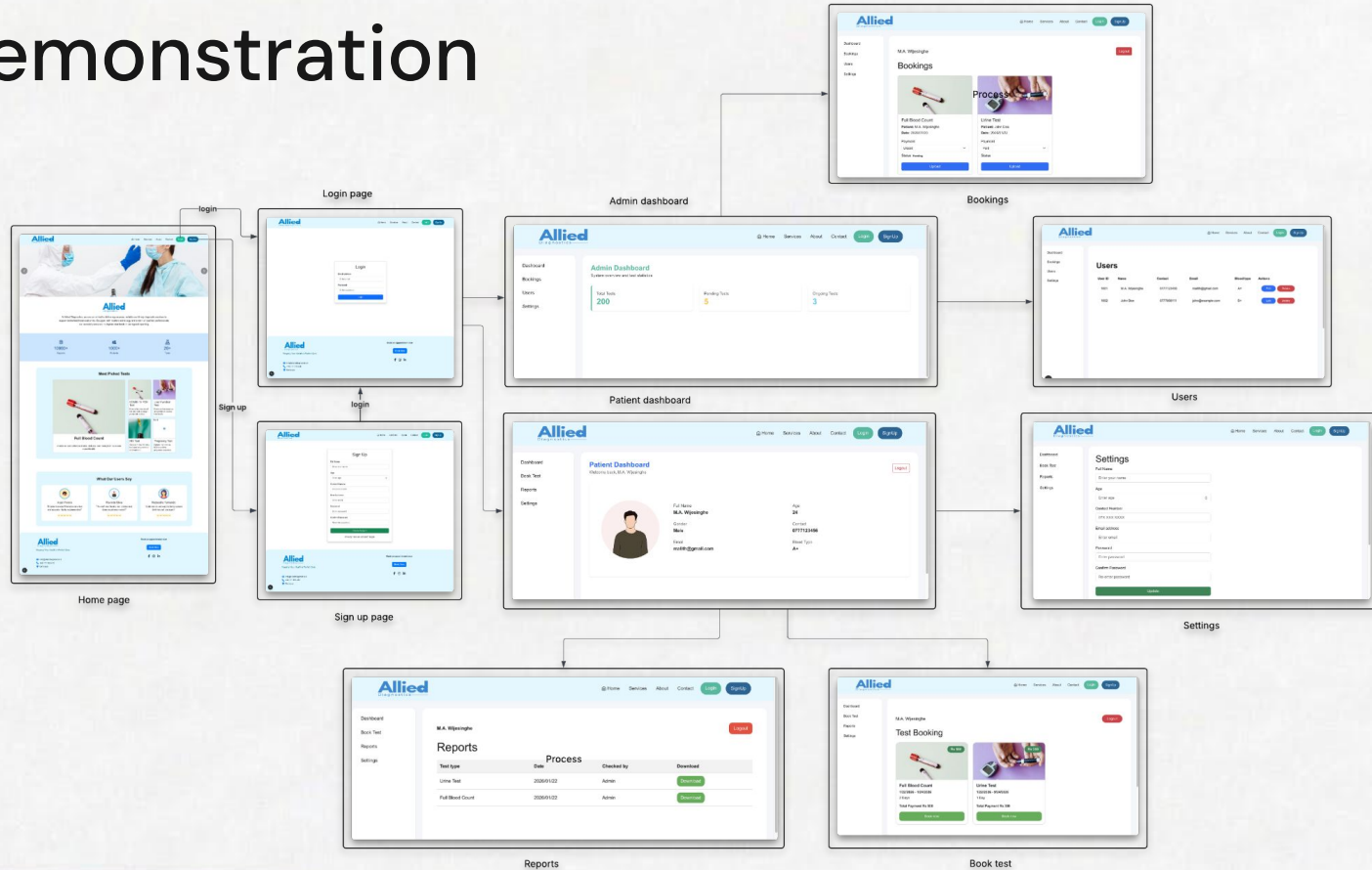
Architecture diagram



Use case diagram

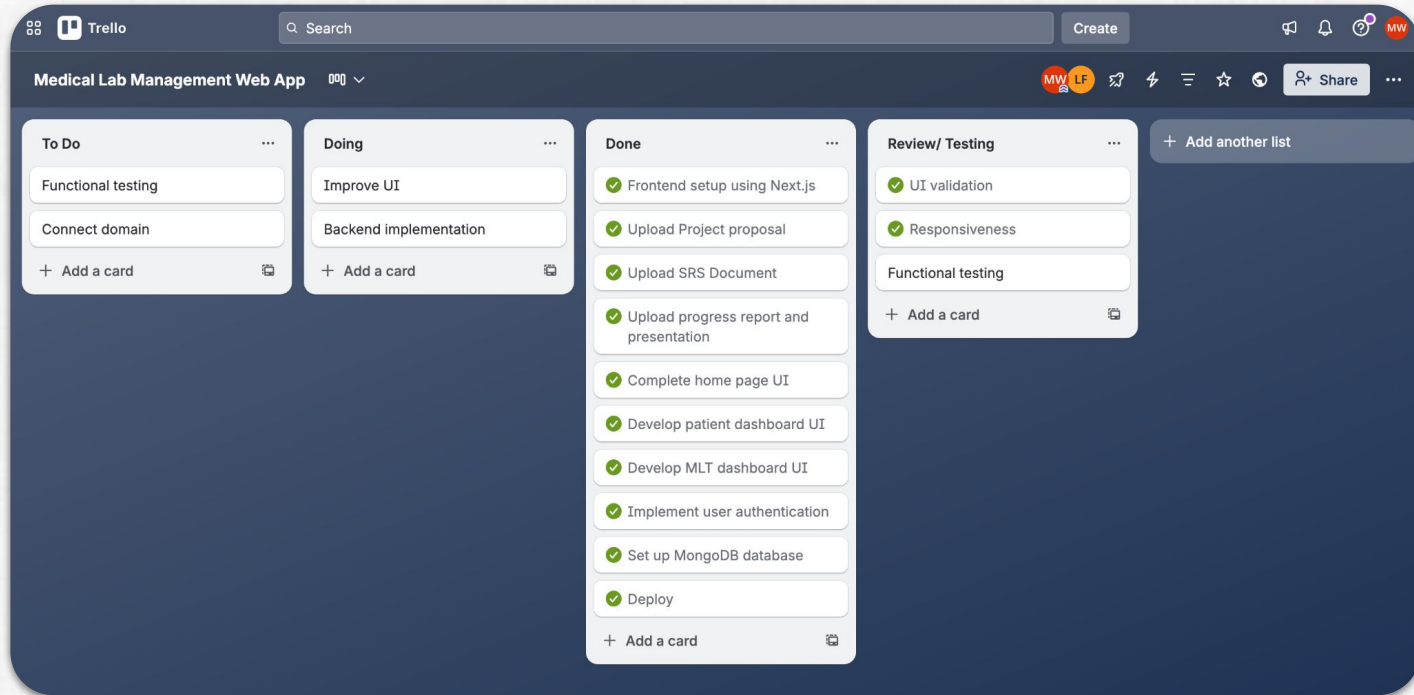


Demonstration



Trello board

 Trello board



The screenshot displays a Trello board for a project named "Medical Lab Management Web App". The board is organized into four columns: "To Do", "Doing", "Done", and "Review/ Testing".

- To Do:**
 - Functional testing
 - Connect domain
 - + Add a card
- Doing:**
 - Improve UI
 - Backend implementation
 - + Add a card
- Done:**
 - ✓ Frontend setup using Next.js
 - ✓ Upload Project proposal
 - ✓ Upload SRS Document
 - ✓ Upload progress report and presentation
 - ✓ Complete home page UI
 - ✓ Develop patient dashboard UI
 - ✓ Develop MLT dashboard UI
 - ✓ Implement user authentication
 - ✓ Set up MongoDB database
 - ✓ Deploy
 - + Add a card
- Review/ Testing:**
 - ✓ UI validation
 - ✓ Responsiveness
 - Functional testing
 - + Add a card

The interface includes a top navigation bar with a search bar, a "Create" button, and user avatars. The board title "Medical Lab Management Web App" is visible at the top left of the workspace. A "Share" button is located at the top right of the board area.

git

 Live repository

  OUSL-EEY4189-WALP-3 / lab-management-system

[Code](#) [Issues](#) [Pull requests](#) [Actions](#) [Projects](#) [Wiki](#) [Security](#) [Insights](#) [Settings](#)

 **lab-management-system** Public

[Edit Pins](#) [Watch 0](#) [Fork 0](#) [Star 0](#)

[main](#) [7 Branches](#) [0 Tags](#)

[Add file](#) [Code](#)

 **MalithWijesinghe** Merge pull request #6 from OUSL-EEY4189-WALP-3/style/dash... 01a8436 · 2 months ago 20 Commits

 public	Added user reviews section and fixed layout issuer	5 months ago
 src	Fixing UI Alignment issues	2 months ago
 styles	Fixing UI Alignment issues	2 months ago
 .gitignore	Setup Next.js project with Bootstrap	5 months ago
 .prettierrc	feat: add login and signup pages	3 months ago
 README.md	Add README with project setup instructions	5 months ago
 eslint.config.mjs	Setup Next.js project with Bootstrap	5 months ago
 next.config.ts	Setup Next.js project with Bootstrap	5 months ago
 package-lock.json	Completed homepage layout	5 months ago
 package.json	Initial commit with bootstrap intergration	5 months ago
 tsconfig.json	Initial commit with bootstrap intergration	5 months ago

About

Allied Diagnostic Laboratory Management System

-  Readme
-  Activity
-  Custom properties
-  0 stars
-  0 watching
-  0 forks
-  Audit log
- Report repository

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Challenges Encountered



Time Management Constraints



Limited Experience with Technologies



Learning New Tools and Technologies



Design Decision Conflicts



Limited Team Resources

Final thoughts

1. Project impact

- Improved lab resource management
- Reduced manual record keeping
- Centralized lab information access



2. Lessons learned

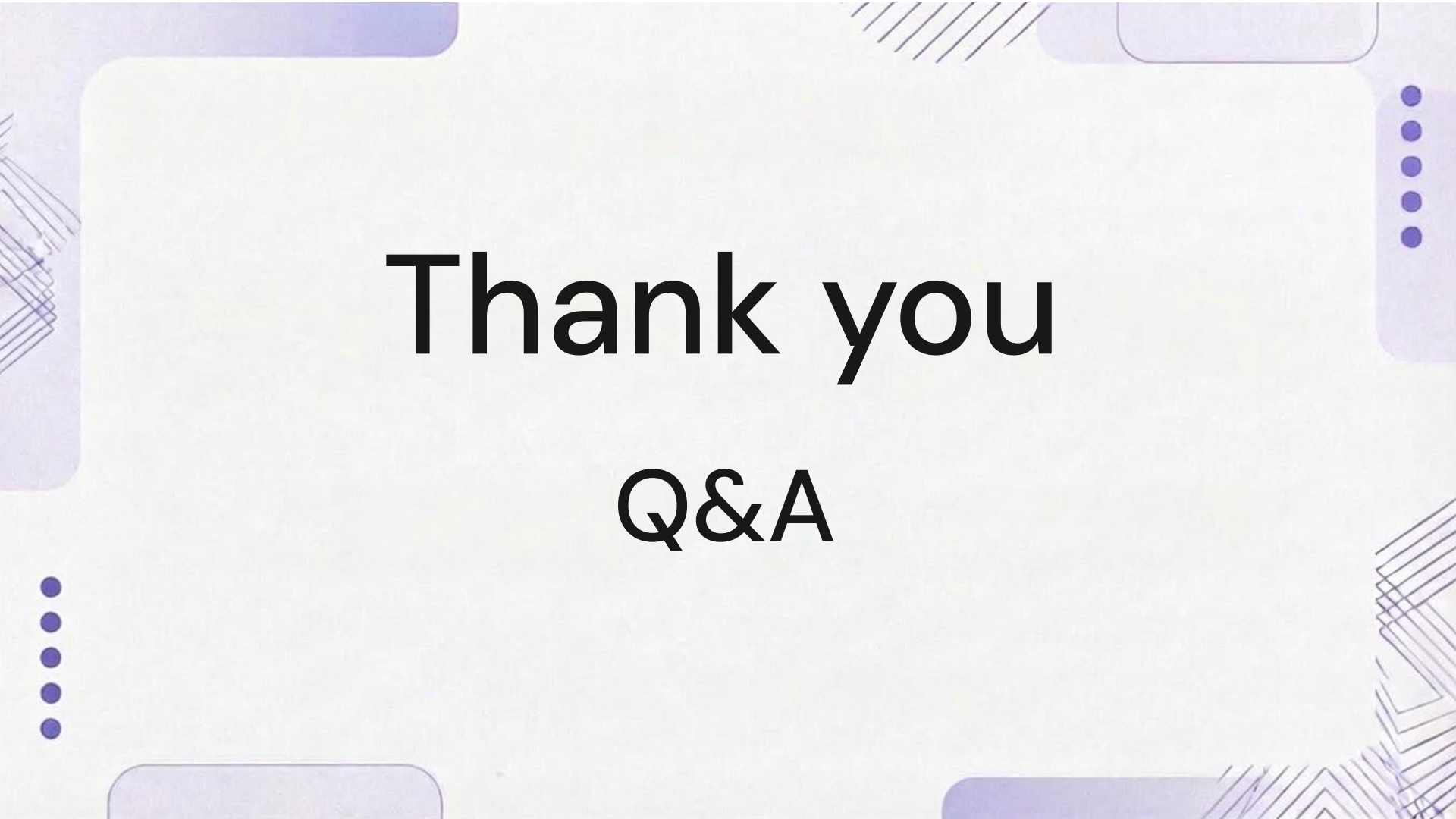
- Gained experience with React and Next.js
- Improved teamwork and communication
- Learned real-world web development



3. Future focus

- Add inventory tracking features
- Develop mobile app
- Improve system security and performance





Thank you

Q&A